

1) Solve the equation $\sqrt{3x-2} = x$

a) $(\sqrt{3x-2})^2 = \underline{\hspace{2cm}} ?$

b) Square both sides of the equation $\sqrt{3x-2} = x$. $x^2 = \underline{\hspace{2cm}} ?$

$$(\sqrt{3x-2})^2 = (x)^2.$$

$$x^2 = \underline{\hspace{2cm}} ?$$

c) Make the right side of the result in part (b) equal to zero and solve for x . $x = \underline{\hspace{2cm}} ?$

d) Graph the function $y = \sqrt{3x-2} - x$. x -intercepts are $\underline{\hspace{2cm}} ?$

TI-84 input: $Y_1 = \sqrt{(3X - 2)} - X$

2) Solve the equation $\sqrt{4x-3}=x$

a) $(\sqrt{4x-3})^2 = \underline{\hspace{2cm}}?$

b) Square both sides of the equation $\sqrt{4x-3}=x$.

$$(\sqrt{4x-3})^2 = (x)^2.$$

$$x^2 = \underline{\hspace{2cm}}?$$

c) Make the right side of the result in part (b) equal to zero and solve for x . $x = \underline{\hspace{2cm}}?$

d) Graph the function $y = \sqrt{4x-3} - x$. x -intercepts are $\underline{\hspace{2cm}}?$

TI-84 input: $Y_1 = \sqrt{\hspace{1cm}}(4X - 3) - X$

3) Solve the equation $\sqrt{-2x-1} = x+2$

a) $(\sqrt{-2x-1})^2 = \underline{\hspace{2cm}}?$

b) Square both sides of the equation $\sqrt{-2x-1} = x+2$.

$$(\sqrt{-2x-1})^2 = (x+2)^2.$$

$$x^2 + 4x + 4 = \underline{\hspace{2cm}}?$$

c) Make the right side of the result in part (b) equal to zero and solve for x . $x = \underline{\hspace{2cm}}?$

d) Graph the function $y = \sqrt{4x-3} - x$. x -intercepts are $\underline{\hspace{2cm}}?$

TI-84 input: $Y_1 = \sqrt{-2X - 1} - X - 2$

4) Solve the equation $\sqrt{2x+2} + 3 = x$

Hint: Subtract 3 from both sides of equation. $\sqrt{2x+2} + 3 = x \Rightarrow \sqrt{2x+2} = x - 3$

a) $(\sqrt{2x+2})^2 = \underline{\hspace{2cm}} ?$

b) Square both sides of the equation $\sqrt{2x+2} = x - 3$.

$$(\sqrt{2x+2})^2 = (x-3)^2.$$

$$x^2 - 6x + 9 = \underline{\hspace{2cm}} ?$$

c) Make the right side of the result in part (b) equal to zero and solve for x . $x = \underline{\hspace{2cm}} ?$

d) Graph the function $y = \sqrt{2x+2} + 3 - x$. x -intercepts are $\underline{\hspace{2cm}} ?$

TI-84 input: $Y_1 = \sqrt{(2X+2)} + 3 - X$

